

AI & Machine Learning - Task 2 Report

Feature Engineering, Model Optimization & Performance Comparison using the California Housing Dataset.

Objective

Build and compare regression models after preprocessing and feature engineering.

Dataset

California Housing Dataset from scikit-learn.

Methodology

Load data, inspect, engineer features, split dataset, scale features, train Linear Regression, Ridge Regression and Decision Tree Regressor, then compare RMSE and R².

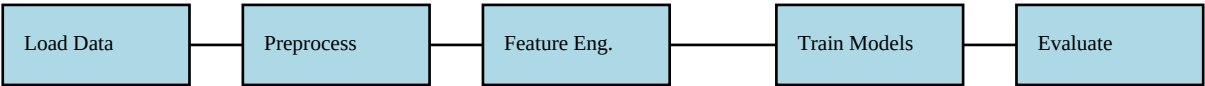
Feature Engineering

Created Rooms_per_Household, Bedrooms_per_Room, Population_per_Household.

Evaluation

RMSE and R² Score were used to compare models.

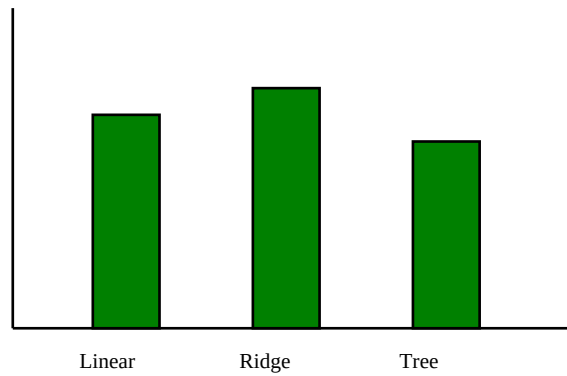
Workflow Diagram



Performance Comparison Table

Model	RMSE	R ²
Linear Regression	(Run notebook)	(Run notebook)
Ridge Regression	(Run notebook)	(Run notebook)
Decision Tree	(Run notebook)	(Run notebook)

Illustrative Model Comparison Diagram



Conclusion

The workflow demonstrates preprocessing, feature engineering, regression model training, and performance evaluation. Replace the placeholder RMSE and R^2 values with the actual results from your notebook execution before final submission.